

Predicting the Influence of Music Education on Academic Performance in Children

A Senior Project submitted to
The Division of Science, Mathematics, and Computing
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by
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Abstract

Music education is frequently promoted as a way to improve children’s cognitive development and academic performance. However, prior research on the topic has produced mixed findings, often due to differences in methods used and the underlying role of socioeconomic status. This project examines the relationship between music education and academic performance using data from the Panel Study of Income Dynamics Child Development Supplement (PSID-CDS) for 2014, 2019, and 2021. After data cleaning and normalization, 1,173 students remained in the sample. Statistical methods included descriptive analysis, t-tests, correlations, regression, and predictive modeling.

Results show that direct measures of music participation – taking lessons or general music involvement – were weak and inconsistent predictors of GPA. In contrast, having a musical instrument at home was the most stable and statistically significant predictor across all three years. The combined regression model explained 5.9% of the variance in GPA ($R^2 = 0.059$), suggesting that academic performance is influenced by factors beyond music variables. Predictive models also demonstrated limited explanatory power, reinforcing the complexity of academic performance.

The findings suggest that music education contributes to student success indirectly, through broader factors like household and environmental influences, rather than as a strong independent cause of higher academic performance. This project supports previous findings that music education is a nuanced component of child development, rather than a standalone determinant of academic performance and outcomes.

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Dedication

To my Dad - Borislav, my Mom - Lidia, my Brother - Radoslav, my Grandma, and my entire family, for believing in me and my journey no matter where it led me. To my friends, who were there to listen when nothing made sense. To my mentors, who showed me how to believe in myself. And to my younger self, who never stopped dreaming.

Acknowledgments

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Finally, I would like to thank my family, friends, and mentors for their continued encouragement and support throughout my time at Bard College.

1

Introduction

Music has been a part of human society since the earliest civilizations. It served as a means of communication, ritual, storytelling, celebration, and community identity. Across cultures and historical periods, music has been a social practice that expresses emotion, preserves tradition, and builds connections. As societies developed, music became institutionalized through religious practice, cultural ceremonies, and eventually formal education. Today, music education is a common part of educational systems, private teaching, and extracurricular activities.

Music education is often presented as valuable not only for the development of artistic qualities, but also for its broader benefits to children and adolescents. Parents, educators, and policymakers often argue that music lessons improve discipline, concentration, memory, creativity, and academic performance. School music programs are often seen as a way to strengthen math skills, language development, executive functioning, and social-emotional growth.

However, the existing scientific evidence for these claims provides mixed results. Some studies link musical training to stronger cognitive performance, improved academic outcomes, and enhanced self-regulation (Barbaroux, Dittinger, & Besson, 2019; Román-Caballero, Vadillo, Trainor, & Lupiáñez, 2022). Others argue that these findings are

overstated, methodologically weak, or influenced by factors like family income, parental involvement, and pre-existing student ability (Sala & Gobet, 2020; Schellenberg, 2006).

The argument that students who participate in music may already differ in ways that contribute to their academic success, compared to those that do not, is central to making the distinction between correlation and causation of music education. If music education directly improves learning outcomes, expanding access to music programs could be an effective way to boost overall academic performance. But if music participation mainly reflects pre-existing advantages, the benefits of music may be less about music itself and more about the environments that make music participation possible.

Socioeconomic status (SES) is crucial in this context. Historically, formal music education was restricted to wealthier families due to the cost of instruments and lessons. Modern technology has lowered some barriers, but access to music education remains uneven. Students from higher-income households are more likely to own instruments, receive private lessons, and participate in music programs. As a result, music education can serve as both a developmental opportunity and a marker of broader social inequality.

This project examines the relationship between music education and academic performance through a large-scale quantitative analysis of children in the United States. Using data from the Panel Study of Income Dynamics Child Development Supplement (PSID-CDS) for 2014, 2019, and 2021, the study investigates whether music participation, access to instruments, and related arts activities are associated with student grade point average (GPA). Unlike smaller studies focused on single schools or local samples, the PSID provides a broader national dataset with repeated survey waves, allowing relationships to be compared across multiple cohorts.

The project incorporates predictive modeling in addition to traditional statistical analysis. This allows us to test whether relationships exist and evaluate the practical explanatory power of music-related variables when predicting academic outcomes.

The central research question and hypothesis of this paper are: *To what extent is music education associated with academic performance, and do music-related variables meaningfully predict student GPA once broader patterns are considered?*

By addressing this question, the project aims to contribute to a more realistic understanding of music education. Rather than assuming music is either a powerful academic intervention or an irrelevant extracurricular activity, the goal is to evaluate where its measurable influence appears, how strong that influence is, and what broader factors may shape it.

1.1 Literature Review

Research on the developmental value of music education has expanded significantly over the past several decades. Scholars in education, psychology, neuroscience, and sociology have examined whether music training contributes to outcomes beyond musical skills. Proposed “non-musical outcomes” include academic achievement, executive functioning, language development, memory, attention, discipline, and emotional growth. While many studies report positive associations between music participation and child development, the size, consistency, and causal interpretation of these effects remain contested.

Early research often relied on descriptive comparisons between students enrolled in music programs and those who were not. Studies from the 1970s found that students participating in solo instrumental music or school ensembles earned higher grades and performed better on standardized assessments than nonparticipants (Wolff, 1978; Anello, 1972). These findings helped establish the idea that music education might contribute to academic success. However, many early studies lacked rigorous controls for prior achievement, family background, or self-selection into music programs, making them more effective at identifying correlations than proving causation.

As research expanded, it began exploring the possibility of music influencing general development. Studying music requires sustained attention, memory, motor coordina-

tion, and repetitive practice. Miendlarzewska and Trost (2014) argue that music engages reward systems and executive processes in ways that may transfer to broader learning contexts. Habibi et al. (2018) reviewed longitudinal evidence, suggesting that sustained music instruction may support auditory processing, self-regulation, and cognitive flexibility. Shen et al. (2019) reported similar improvements in executive functioning among preschool children exposed to structured music training. These studies support the hypothesis that music may strengthen cognitive systems relevant to classroom learning.

Other research focuses directly on academic outcomes. Yang et al. (2014), used longitudinal data to find that children with sustained music training demonstrated stronger academic development over time. Costa-Giomi (2004) examined children who took piano lessons and reported small improvements in school performance and self-esteem. Hallam (2010) reviewed a broad range of studies, concluding that music participation may positively influence intellectual, personal, and social development, while emphasizing that outcomes depend heavily on duration, intensity, and instructional quality.

Despite such findings in support of the hypothesis of music improving performance, recent meta-analyses show a different aspect of it. Sala and Gobet (2020), in one of the most influential critiques of the field, analyzed a large body of studies and concluded that the cognitive and academic benefits of music training were generally small and often disappeared under more rigorous research designs. They argued that randomized controlled trials produced substantially weaker effects than observational studies, suggesting that many previously reported benefits could reflect selection bias rather than causation from music training itself.

Román-Caballero et al. (2022) conducted a comprehensive meta-analysis and concluded that instrumental music training does produce small but statistically significant academic and cognitive benefits. An important part of their research is their distinction between randomized, quasi-randomized, and self-selection studies, noting that method-

ology strongly shapes observed effect sizes. Their findings suggest music education has real benefits. Taken together, these meta-analyses further show the variance in results on the topic. Music education matters, but the magnitude of its effects is often amplified by popular narratives.

Another major theme in this debate is the role of socioeconomic status and its impact. Access to music education is not evenly distributed. Lower-income families struggle to afford private lessons, instruments, or long-term extracurricular programs (Beveridge, 2022). Albert (2006) similarly argued that instrumental participation is strongly shaped by resources. Bates (2023) draws on Bourdieu’s concept of cultural capital, suggesting that access to music education may reproduce social inequality by giving already advantaged children additional developmental opportunities.

Research suggests music education may be valuable when barriers to access are reduced. Barbaroux, Dittinger, and Besson (2019) studied children from low socioeconomic backgrounds and found that music training positively influenced cognitive functioning. This implies music programs can produce stronger benefits for students who lack opportunities. In this sense, socioeconomic status may function both as a barrier to access and as a moderator of its impact.

Recent studies have expanded beyond academic outcomes to examine psychosocial development. Villanueva, Ilari, and Habibi (2024) found long-term music instruction was associated with emotional skill development. Bussu and Mangiarulo (2024) reported ensemble participation supported adolescents’ perceptions of teamwork, responsibility, and life skills. These findings suggest music education’s value goes beyond GPA or test scores.

The literature does not support a simple yes or no answer on music education’s benefits. It shows a series of broad conclusions. First, music participation is frequently associated with positive developmental outcomes. Second, effect sizes are typically small to moderate rather than significant. Third, socioeconomic context, parental support, and

selection effects are essential for interpreting these relationships. This creates a strong case for large-scale data analysis using nationally recognized datasets. While observational data cannot establish causality, larger samples can help determine which music-related variables show stable relationships with academic performance.

2

Methods

The methodological approach of this project combines data background, preparation, descriptive statistics, inferential analysis, regression modeling, and predictive modeling. The primary outcome variable is normalized GPA, while explanatory variables represent different forms of musical participation, arts involvement, and household access to musical resources.

2.1 Data Background

The data used in this project is from the Panel Study of Income Dynamics (PSID) and its Child Development Supplement (CDS). The PSID is one of the longest-running and most widely respected longitudinal household surveys in the United States. It began in 1968 at the University of Michigan with the purpose of tracking a nationally representative sample of American families over time. As children from original sample households became adults and formed their own households, they and their descendants remained in the study. This multi-generational design makes the PSID especially valuable for research involving education, income mobility, family structure, and long-term social outcomes.

Since its establishment the PSID has collected extensive information on employment, earnings, wealth, health, demographic characteristics, housing conditions, educa-

tion, and family relationships. Because the same families are followed repeatedly, the dataset allows researchers to examine how childhood environments relate to the outcomes of socioeconomic status and how social advantages or disadvantages persist across generations.

To better study child and adolescent development specifically, the PSID introduced the Child Development Supplement (CDS) in 1997. It focuses on children living in PSID households and includes detailed measures of schooling, learning environments, time use, extracurricular activities, health, family interactions, and developmental outcomes. It is particularly useful for educational research because it includes variables related to academic performance and household enrichment opportunities.

This project uses CDS data from 2014, 2019, and 2021. These three survey waves were selected because they contain comparable measures related to music education, arts participation, household musical resources, and grade point average. Using multiple years strengthens the study in several ways. It increases total sample size and it allows the consistency of findings to be tested across cohorts rather than relying on one isolated year. It also provides a time-series perspective on whether established patterns remain stable over time.

One of the major strengths of using PSID-CDS data is the possibility of combining datasets. Many studies of music education rely on single schools, regional districts, or convenience samples. By contrast, the PSID includes respondents from varied geographic, racial, and socioeconomic backgrounds across the United States. This gives the project a stronger basis for evaluating whether relationships between music-related variables and academic outcomes appear consistently across broader populations.

At the same time, there are several limitations of the data. GPA is self-reported rather than verified through transcripts and some participation variables are binary rather than ordinal. As a result, the study can identify associations but cannot definitively establish causal effects. Despite these limitations, the PSID Child Development Supplement

remains one of the strongest publicly available datasets for studying how family environments, extracurricular engagement, and household resources relate to children’s educational outcomes. Its broad scope, repeated survey waves, and inclusion of music-related variables make it particularly well suited for this project.

2.2 Data Preparation and Cleaning

The first step of the analysis involved importing the PSID-CDS datasets for 2014, 2019, and 2021 into Python using the `pandas` library. Because variable names differ across survey waves, all variables were renamed into a consistent format to allow for comparison and merging.

Only variables directly relevant to the research question were retained. These included:

- music lessons in the past year
- general music participation
- presence of a musical instrument in the home
- frequency of instrument use
- frequency of use of electronics for music listening
- arts tutoring
- arts community participation
- arts participation in religious settings
- raw GPA
- GPA scale

Observations with missing GPA values or missing GPA scale values were removed. Cases with a GPA scale of zero were excluded to prevent invalid normalization.

2.3 GPA Standardization

Because grading systems differ across students, raw GPA values were standardized using:

$$GPA_{\text{norm}} = \frac{GPA_{\text{raw}}}{GPA_{\text{scale}}} \quad (2.3.1)$$

This transformation places all GPA values on a 0–1 scale, where 0 represents the lowest possible GPA and 1 represents the highest possible GPA. Any values exceeding 1 were treated as outliers and removed.

2.4 Combined Dataset Construction

After cleaning each dataset individually, the three survey waves were merged into a single combined dataset. A year indicator variable was added to allow comparisons across cohorts. The final dataset included 1,173 valid GPA observations, while regression models used 1,164 observations after removing cases with missing predictor values.

2.5 Variables

Binary variables included:

- music lessons
- music participation
- instrument in the home
- arts tutoring
- arts community participation
- arts religious participation

Ordinal variables included:

- frequency of instrument use (1–6 scale)
- frequency of use of electronics for music listening (1–5 scale)

2.6 Descriptive Statistics

Descriptive statistics were calculated for normalized GPA in each survey year, including sample size (N), mean, median, and standard deviation. Histograms were generated to visualize the distribution of GPA values across years (Figures 3.1.1–3.1.3).

2.7 Independent Samples t-tests

Independent samples t-tests were conducted separately for each survey year to compare mean GPA between students who participated in specific activities and those who did not. These tests were appropriate because GPA is a continuous variable and participation variables are binary. Statistical significance was evaluated using a threshold of $p < 0.05$.

2.8 Point-Biserial Correlations

Point-biserial correlation coefficients were calculated to measure the relationship between binary participation variables and GPA. Correlation strength was interpreted using standard thresholds, with $|r| \approx 0.10$ considered small, $|r| \approx 0.30$ moderate, and $|r| \approx 0.50$ large.

2.9 Multivariable Regression

An Ordinary Least Squares (OLS) regression model was estimated using the combined dataset. OLS regression is a statistical method that estimates the linear relationship between a dependent variable and multiple independent variables by minimizing the sum

of squared differences between observed and predicted values. This approach allows for the evaluation of the unique contribution of each variable while holding all other variables constant.

The dependent variable was normalized GPA, and predictors included all music-related variables as well as survey year. This approach estimates the unique contribution of each variable while controlling for all others.

Reported statistics included coefficients, standard errors, t-values, p-values, confidence intervals, and model fit measures such as R^2 and adjusted R^2 .

2.10 Predictive Modeling

Predictive models were implemented using linear regression from `scikit-learn`. Data was split into training (80%) and testing (20%) sets. Models were estimated separately for each year and for the combined dataset.

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 score

Scatter plots of predicted versus actual GPA values were used to visualize model performance (Figure 3.6.1).

2.11 Software

All of the analyses were conducted in Python using the following libraries:

- `pandas` for data handling
- `numpy` for numerical operations

- `scipy.stats` for statistical tests
- `statsmodels` for regression analysis
- `scikit-learn` for predictive modeling
- `matplotlib` for visualization

3

Results

The results of the statistical analysis are divided into four separate stages. The first stage presents descriptive statistics, summarizing GPA distributions across the three survey years. The second stage used independent samples t-tests to examine the differences in GPA across participation variables. Next, ordinal distributions and point-biserial correlations examine the patterns of engagement and their relationship to academic performance. Finally, regression and predictive modeling results are used to evaluate the combined explanatory and predictive strength of the variables.

3.1 GPA Distributions

Upon conducting the descriptive statistics for the datasets, the results indicate that GPA distribution remains very stable across the different years of sampling, with only a slight upward shift over time (Table 3.1.1). Mean GPA increases from 0.780 in 2014 to 0.809 in 2021, with the combined mean of the three datasets being 0.793. Standard deviations are distributed between 0.142 and 0.153 across all years, indicating consistent dispersion and no evidence of substantial changes in the overall academic performance of the participants.

Table 3.1.1: Summary statistics of normalized GPA by year

Year	N	Mean	Median	Std. Dev
2014	397	0.780	0.785	0.148
2019	469	0.793	0.820	0.153
2021	305	0.809	0.820	0.142

The distributions of GPA values for each year are shown in Figures 3.1.1–3.1.3. All three distributions are left-skewed, with most students clustering near the upper bound of the GPA scale. This pattern is expected given the bounded nature of GPA.

The 2014 dataset shows the general distributional structure (Figure 3.1.1). The mean normalized GPA is 0.781, the median is 0.786, and the standard deviation is 0.149.

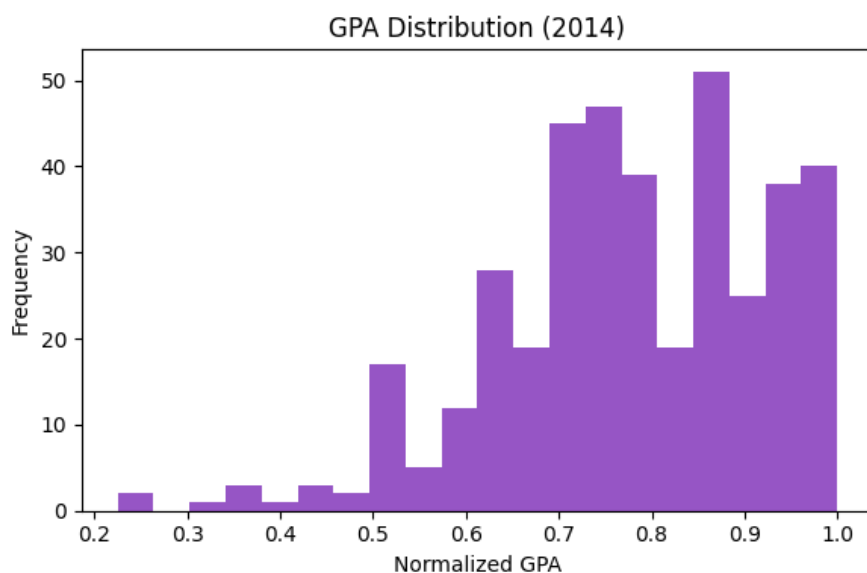


Figure 3.1.1: Distribution of normalized GPA, 2014

The 2019 dataset follows the same pattern (Figure 3.1.2). The mean normalized GPA is 0.793, the median is 0.820, and the standard deviation is 0.153. The slightly higher mean GPA compared to 2014 suggests a small variation in academic performance between different years.

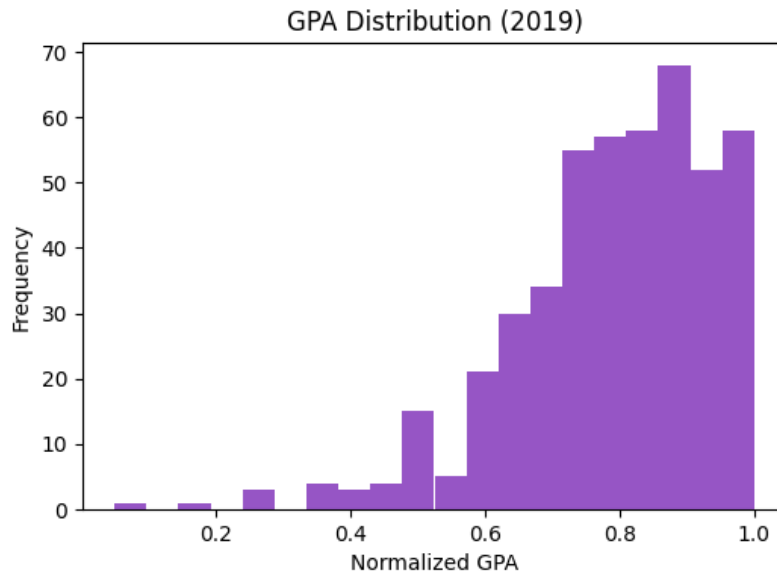


Figure 3.1.2: Distribution of normalized GPA, 2019

In the 2021 dataset, the normalized GPA values produce a mean of 0.809, a median of 0.820, and a standard deviation of 0.142 (Figure 3.1.3). Compared to the earlier datasets, the slightly higher mean and median values could suggest an upward shift in GPA over time, though the overall structure of the distribution remains highly consistent across the years. This pattern is consistent with typical academic grading distributions, where the maximum possible GPA constrains the upper limit of the distribution while allowing a longer lower tail.

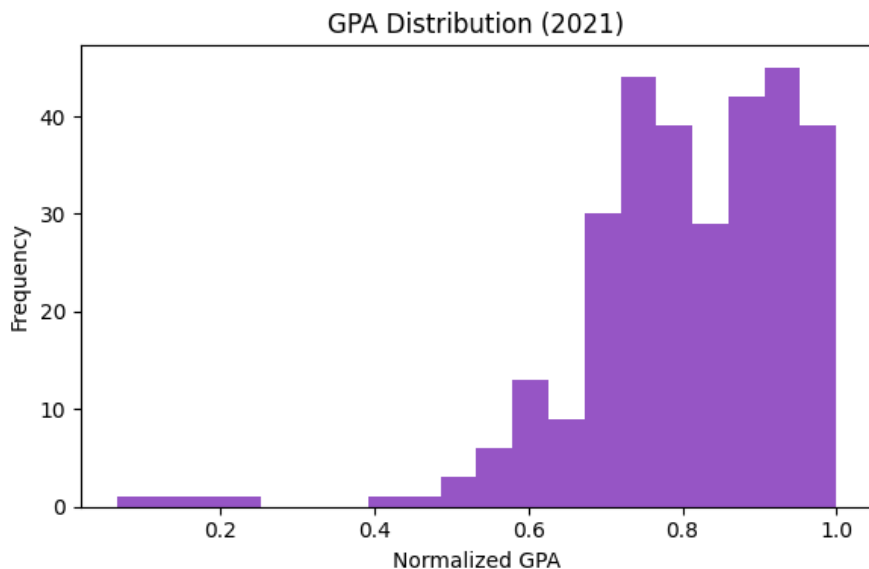


Figure 3.1.3: Distribution of normalized GPA, 2021

Taken together, the GPA distributions remain consistent. Most students fall within the normalized GPA range of approximately 0.75 to 0.90, which corresponds roughly to letter grades between B- and A- on a traditional four-point scale. The stability of the standard deviation between all three years also indicates that the overall variability in academic performance remains relatively consistent across cohorts.

3.2 Statistical Analysis of Binary Variables

After the analysis of the distribution of GPA values, the next step of the analysis was to determine whether students who participated in specific music-related or arts-related activities exhibited statistically significant differences in academic performance compared to students who did not. This involved conducting a series of independent samples t-tests.

The t-tests were applied to several binary variables: participation in music lessons, general music participation, the presence of a musical instrument in the home, participation in arts tutoring, involvement in community arts groups, and participation in

arts-related activities within religious settings, to determine if any of these variables contributed to statistical differences in GPA for the children involved in the dataset (Table 3.2.1).

Table 3.2.1: Independent samples t-tests by year

Variable	2014		2019		2021	
	t	p	t	p	t	p
Music Lessons	2.109	0.036	1.554	0.121	1.498	0.135
Music Participation	0.479	0.633	-0.882	0.380	1.758	0.085
Instrument at Home	4.166	<0.001	2.917	0.004	3.478	0.001
Arts Tutoring	0.648	0.519	—	—	—	—
Arts Community Groups	2.028	0.045	-0.327	0.744	-0.221	0.826
Arts Religious	-0.376	0.708	0.526	0.600	-1.410	0.167

The most consistent t-test result across all three datasets was the presence of a musical instrument in the home. The 2014 dataset produced a statistically significant result with $t = 4.166$ and $p < 0.001$. The 2019 and 2021 datasets produced a similar result with $t = 2.917$, $p = 0.004$ and $t = 3.478$, $p = 0.001$, respectively (Figures 3.2.1–3.2.3). The consistency of these findings across multiple cohorts strongly suggests that the presence of a musical instrument in the household is meaningfully associated with differences in GPA.

Figures 3.2.1, 3.2.2, and 3.2.3 show the GPA distributions for students with and without an instrument at home. Across all three years, students with an instrument at home tend to have higher median GPA values.

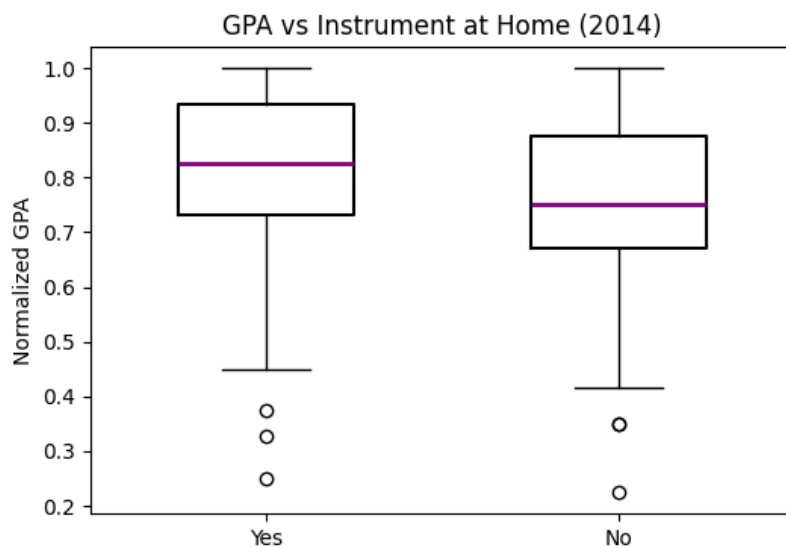


Figure 3.2.1: GPA by instrument availability in the home, 2014

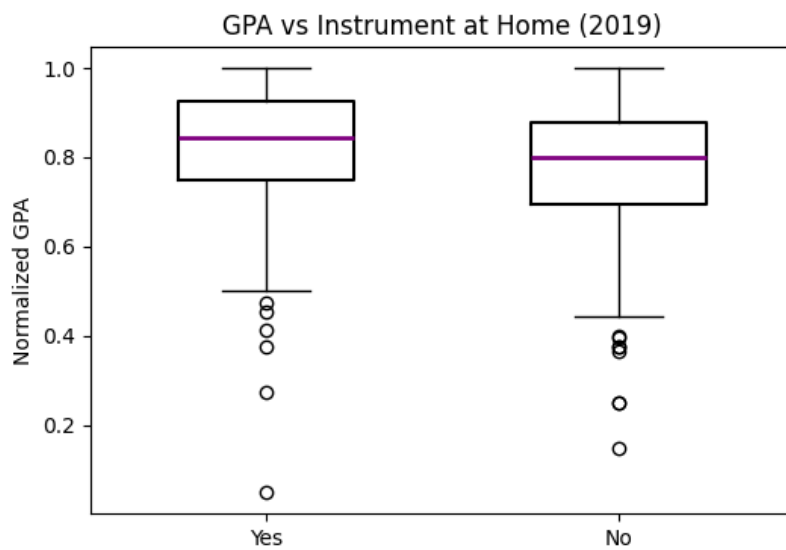


Figure 3.2.2: GPA by instrument availability in the home. 2019

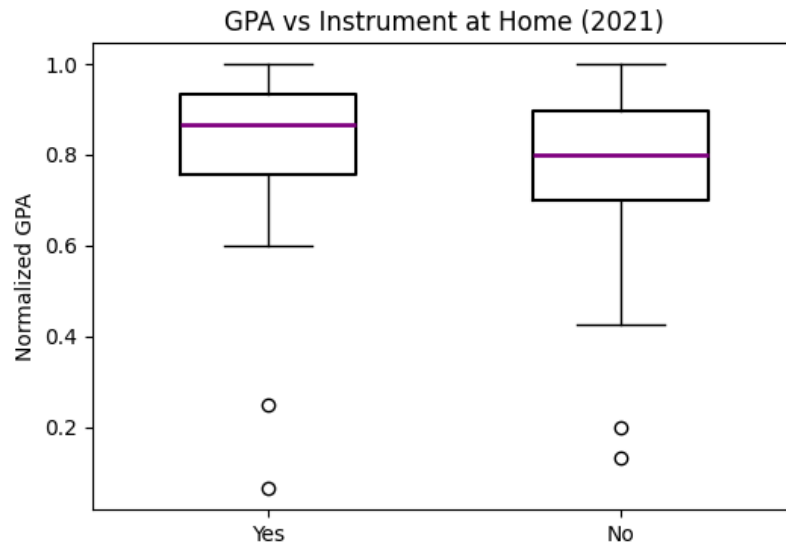


Figure 3.2.3: GPA by instrument availability in the home, 2021

To contrast with the consistent effect of instrument availability, Figures 3.2.4–3.2.6 show GPA differences by music lesson participation. Unlike instrument ownership, these differences are less consistent across years.

The 2014 dataset produced a statistically significant result with $t = 2.109$ and $p = 0.036$. However, in the 2019 dataset, the t-test produced $t = 1.554$ and $p = 0.121$, indicating no statistical significance, which was followed by similar results in the 2021 dataset ($t = 1.498$ with $p = 0.135$). This inconsistency suggests that the relationship between music lessons and academic performance may vary across class years or may depend on additional factors outside of the binary variable alone.

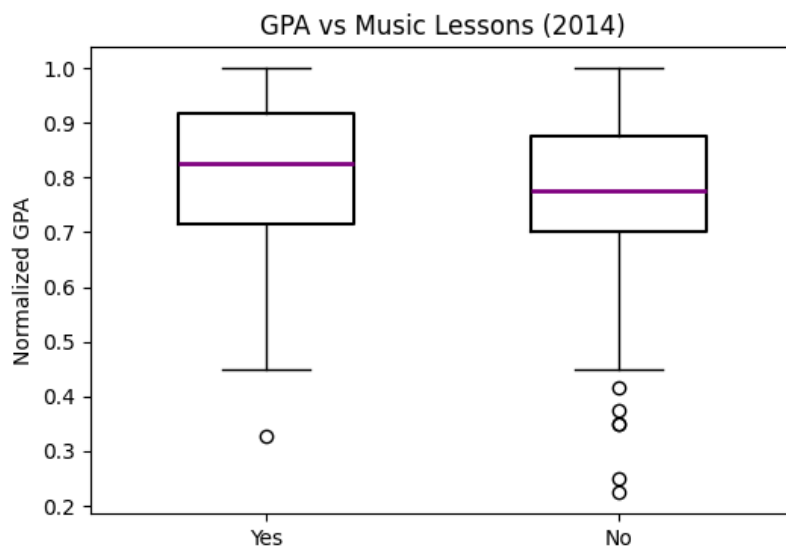


Figure 3.2.4: GPA by music lesson participation in 2014

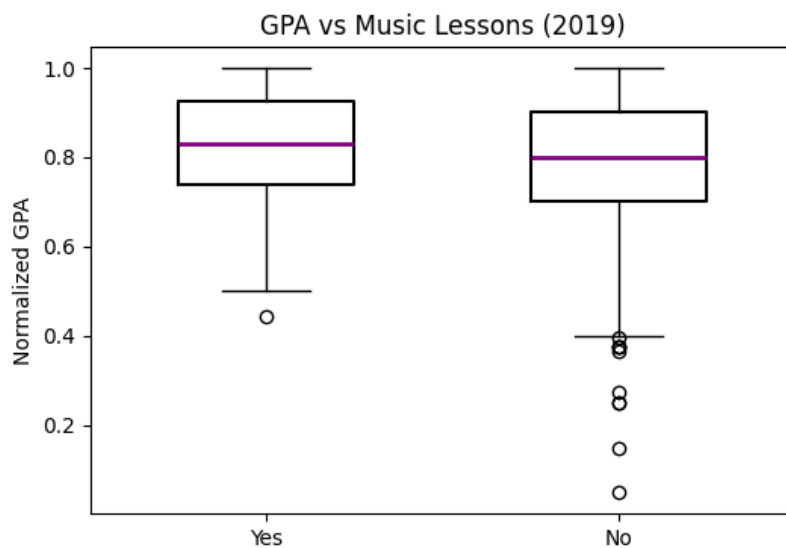


Figure 3.2.5: GPA by music lesson participation in 2019

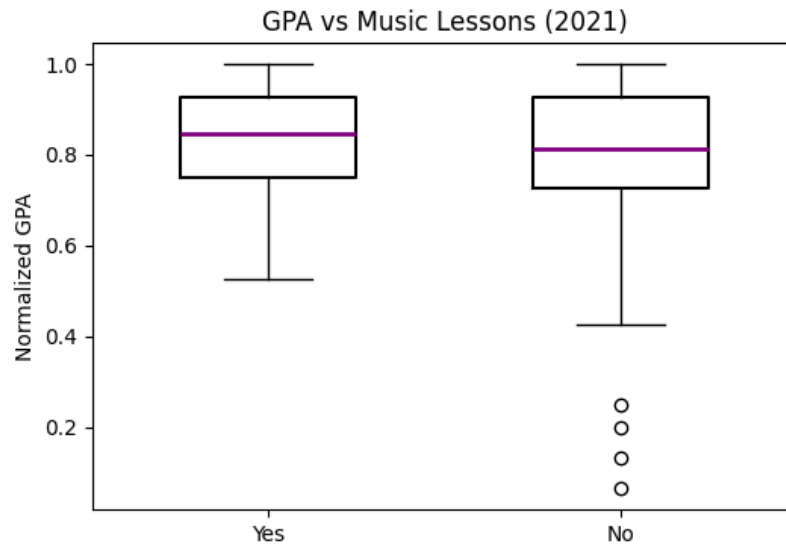


Figure 3.2.6: GPA by music lesson participation in 2021

The overall result of the independent samples t-tests suggest that simple binary indicators of musical participation are not consistently strong predictors of academic performance. However, socioeconomic related factors, such as access to a musical instrument, appear to have a more consistent relationship with GPA outcomes.

Additional comparisons for music lessons and participation are provided in Appendix Figures A.2.1–A.2.3 and A.3.1–A.3.3.

3.3 Statistical Analysis of Ordinal Variables

Two ordinal variables provide information on the intensity of participation: frequency of use of electronics for music listening and frequency of instrument use.

The ordinal variables provide additional information about the frequency and intensity of music engagement. Figure 3.3.1 shows that electronic music listening is highly concentrated in the most frequent response category across all years. This suggests that listening to music electronically is nearly universal among students and may have limited usefulness for differentiating academic outcomes.

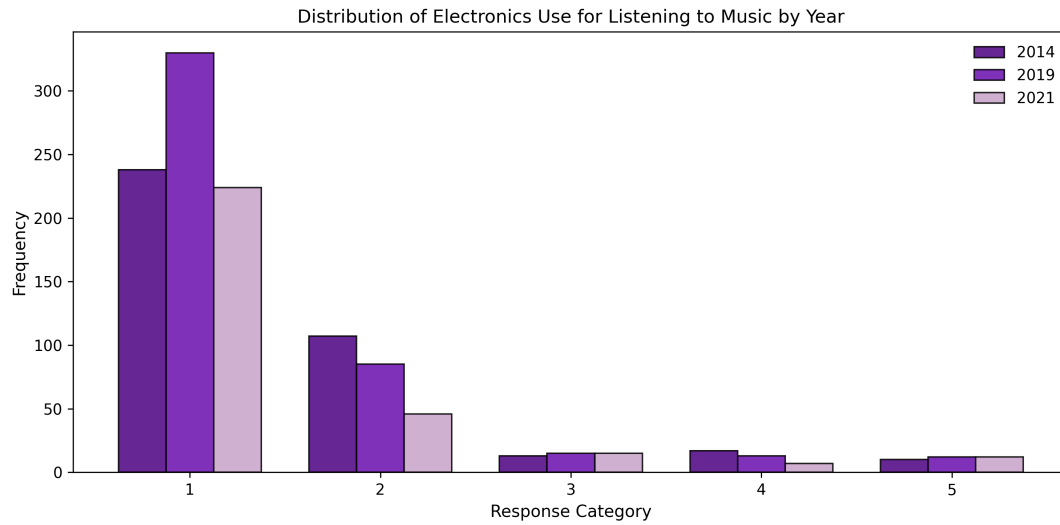


Figure 3.3.1: Distribution of electronics use for listening to music by year

In contrast, Figure 3.3.2 shows greater variation in instrument use frequency. This variable better captures differences in active music engagement because students are more spread across response categories.

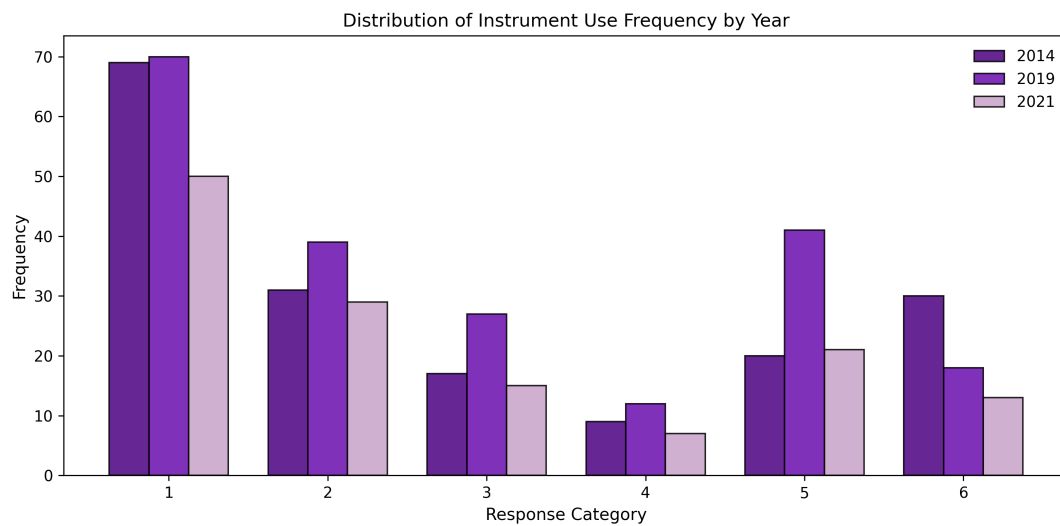


Figure 3.3.2: Distribution of instrument use frequency by year

3.4 Statistical Analysis of the Association between Variables

To further deepen the understanding of the relationship between musical engagement and academic performance, point-biserial correlation coefficients were calculated to determine whether there were any strong associations between variables and their direction. This method is appropriate when one variable is continuous (GPA) and the other variable is binary (participation vs. non-participation) (Table 3.4.1).

Table 3.4.1: Point-biserial correlations with GPA by year

Variable	2014		2019		2021	
	r	p	r	p	r	p
Music Lessons	-0.100	0.046	-0.072	0.121	-0.086	0.135
Music Participation	0.061	0.230	0.078	0.094	0.012	0.837
Instrument at Home	-0.207	<0.001	-0.135	0.004	-0.196	<0.001
Arts Tutoring	-0.119	0.018	-0.134	0.004	-0.062	0.277
Arts Community	0.090	0.074	0.150	0.001	0.044	0.448
Arts Religious	0.128	0.011	0.071	0.126	0.141	0.014

In all datasets, correlations are small in magnitude (generally $|r| < 0.20$), indicating weak relationships. Figures 3.4.1 and 3.4.2 visualize the correlation results by separating negative and positive relationships. This separation makes it easier to observe that most relationships are small in magnitude, even when statistically significant.

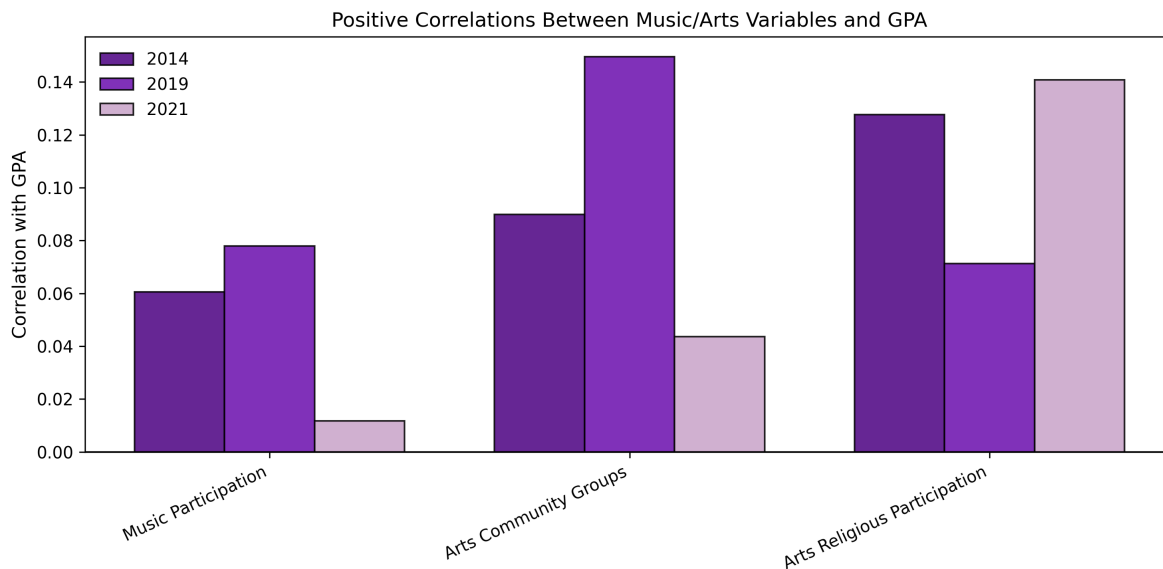


Figure 3.4.1: Positive correlations between music and arts variables and GPA

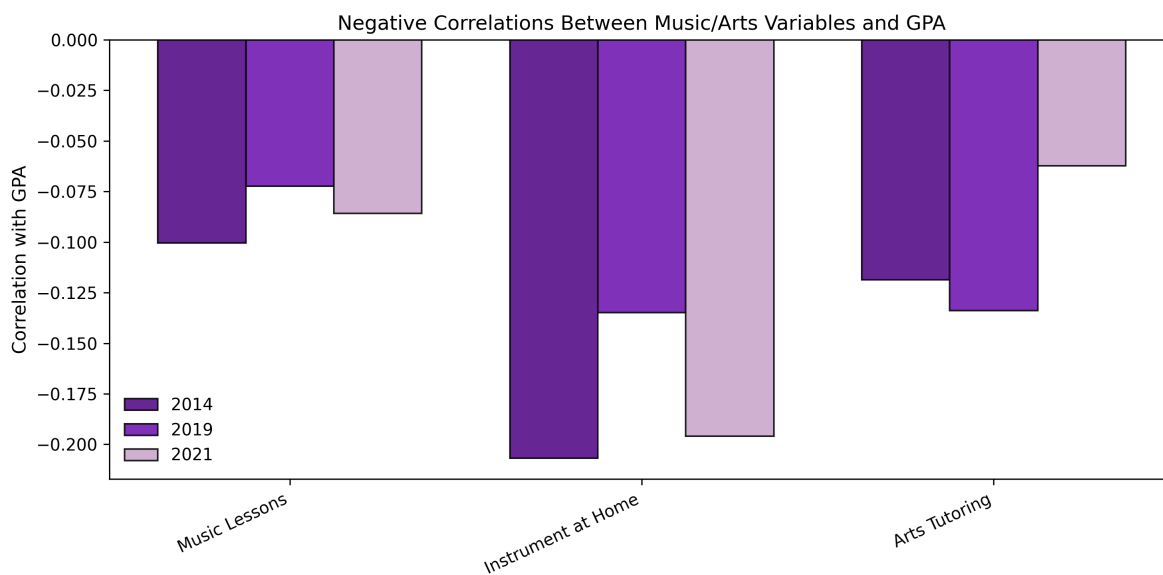


Figure 3.4.2: Negative correlations between music and arts variables and GPA

In 2014, the presence of a musical instrument in the home is associated with a correlation coefficient of $r = -0.207$ with $p < 0.001$. A similar relationship appears in 2019, with $r = -0.135$ and $p = 0.004$, and again in 2021, where the correlation is $r = -0.196$ with $p < 0.001$. While the negative sign reflects the numerical encoding

of the binary variable rather than a true negative relationship, the results consistently indicate that students with instruments at home tend to have higher GPAs. The stability of this pattern across all three years suggests that instrument availability is one of the most reliable predictors of academic performance among the variables examined.

Other variables produce smaller and less consistent correlations across years. Arts tutoring is statistically significant in 2014 ($r = -0.119$, $p = 0.018$) and 2019 ($r = -0.134$, $p = 0.004$), but not in 2021. Participation in community arts groups is not significant in 2014, becomes significant in 2019 ($r = 0.150$, $p = 0.001$), and is not significant in 2021. Participation in religious arts activities is significant in 2014 ($r = 0.128$, $p = 0.011$) and 2021 ($r = 0.141$, $p = 0.014$), but not in 2019.

These patterns suggest that while certain forms of arts engagement are associated with academic performance, their effects are not stable across cohorts and may depend on additional contextual factors such as family background, socioeconomic status, or school environment.

3.5 Multivariable Regression Analysis

To examine how music-related variables are associated with academic performance when considered simultaneously, an Ordinary Least Squares (OLS) regression model is estimated using the combined dataset.

The results of the pooled regression model are presented in Table 3.5.1.

Table 3.5.1: OLS regression results (combined dataset)

Predictor	Coefficient	p-value
Music Lessons	-0.0016	0.730
Music Participation	0.0034	0.501
Instrument at Home	-0.0096	0.003
Instrument Frequency	0.0014	0.726
Electronics Music	0.0009	0.844
Arts Tutoring	-0.0093	<0.001
Arts Community	0.0041	0.046
Arts Religious	0.0061	0.009
Year	0.0039	0.010

The model explains 5.9% of the variance in GPA ($R^2 = 0.059$), indicating that the included variables capture only a small portion of overall academic performance. Several predictors are statistically significant, including the presence of a musical instrument in the home, arts tutoring, community arts participation, religious arts participation, and survey year.

The coefficient for instrument availability is negative due to the encoding of the binary variable rather than indicating a negative relationship. Consistent with earlier results, this finding reflects that students with access to a musical instrument at home tend to have higher GPAs.

Other variables show mixed results. Arts tutoring is associated with slightly lower GPA. Participation in arts in community and religious activities shows small positive associations with academic performance. The year variable is also statistically significant, showing the small upward trend in GPA observed across survey waves.

In contrast, direct measures of music participation, including music lessons and general music involvement, are not statistically significant when controlling for other factors. This suggests that these variables do not independently explain variation in GPA once broader environmental and contextual factors are taken into account.

3.6 Predictive Modeling

While the regression model identifies statistically significant relationships, predictive modeling evaluates whether these variables meaningfully improve the ability to estimate academic performance.

The results of the predictive modeling are summarized in Table 3.6.1.

Table 3.6.1: Predictive model performance by dataset

Dataset	MAE	RMSE	R^2
2014	0.115	0.141	-0.017
2019	0.119	0.154	0.018
2021	0.101	0.122	0.036
Combined	0.112	0.152	0.042

Predictive model performance is weak across all years. The respective models produce low R^2 values, including a negative value in 2014, indicating that the model performs worse than a simple baseline prediction using the mean GPA. Prediction errors remain relatively stable across years, with MAE ranging from approximately 0.10 to 0.12 and RMSE ranging from approximately 0.12 to 0.15, showing minimal improvement in predictive accuracy across specifications.

Predictions cluster near the mean GPA (approximately 0.80), indicating limited ability to distinguish between high and low-performing students. The combined model is shown in Figure 3.6.1, while yearly prediction plots are included in Appendix Figures A.1.1–A.1.3.

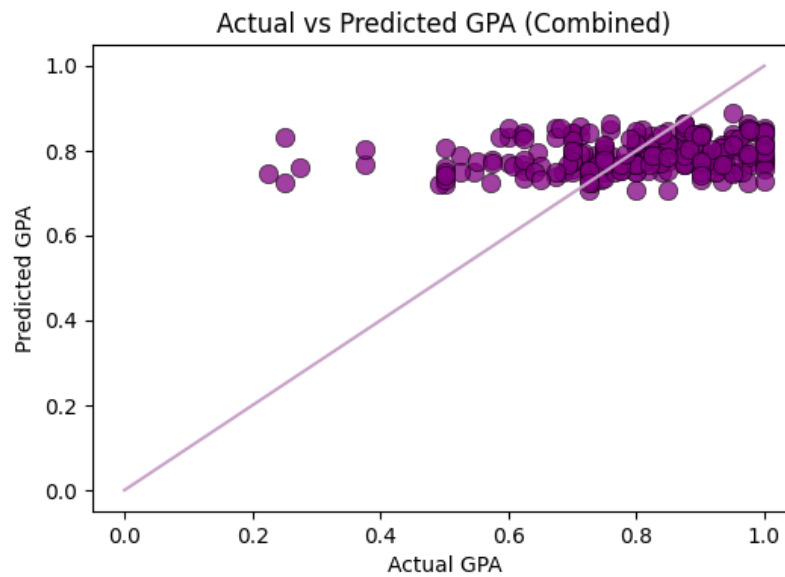


Figure 3.6.1: Predicted versus actual GPA values for the combined dataset

Overall, the results indicate that music-related variables are associated with GPA in small but consistent ways, with environmental factors showing the strongest and most stable relationships.

4

Discussion

The purpose of this project was to investigate whether music education and music-related environmental factors are linked to academic performance in children. Using three waves of the Panel Study of Income Dynamics Child Development Supplement (2014, 2019, and 2021), the study combined descriptive statistics, inferential testing, regression analysis, and predictive modeling to examine these relationships over time. Results showed that while several music-related variables are statistically associated with GPA, the overall magnitude of these relationships is small, indicating a meaningful impact but not a simple one.

The most significant finding of the project was the impact of a musical instrument in the home, which was the most consistent predictor across all analyses and datasets. This variable produced statistically significant t-tests in every year, statistically significant point-biserial correlations in every year, and remained significant in the combined regression model after controlling for other predictors. No other variable demonstrated this level of consistency. This result is meaningful because it suggests that the relationship between music and academic achievement may be influenced more by the broader environment in which a child develops than by participation alone. A household with musical instruments may reflect multiple overlapping advantages, such as greater parental

support, stronger investment in education, higher household resources, or the encouragement of extracurricular learning. In this sense, the instrument variable likely functions as both a direct and indirect measure. It may capture actual musical opportunity while simultaneously representing forms of socioeconomic capital.

This interpretation aligns closely with the literature reviewed earlier. Researchers like Beveridge (2022) and Bates (2023) argue that access to music education is unevenly distributed and strongly shaped by socioeconomic status. Families with greater financial resources are more able to purchase instruments, fund lessons, and secure support for sustained music-related activities. Therefore, the significance of instruments in the home does not necessarily imply that ownership causes higher GPA, but rather indicates conditions associated with academic success.

In contrast, direct participation variables like music lessons and general music participation were inconsistent predictors. Music lessons were significant only in the 2014 sample and non-significant in 2019 and 2021. Music participation was non-significant in all three yearly t-tests and also non-significant in the combined regression model. This suggests that simple yes/no indicators of participation may not capture the complexity of music education. There are several possible reasons for this. Binary variables cannot accurately capture the intensity of training and participation. For example, a student who took one month of lessons would be grouped with a student who took lessons for five years. Similarly, when looking at overall music participation, a child who participated in school chorus would be grouped with a child who practiced for hours daily. These forms of participation differ considerably in intensity, duration, and impact.

Another factor that could be considered is the effect of self-selection. Students who choose music activities may differ in motivation, family structure, or education quality. These are factors that aren't fully captured by a binary variable.

The weak performance of direct participation variables echoes prior research. Meta-analyses like Román-Caballero et al. (2022) report small positive effects of mu-

sic training, while Sala and Gobet (2020) argue that many effects shrink under stronger controls. The findings in this project support an interpretation that music education has benefits, but those benefits are generally modest and often shaped by external factors.

Arts participation outside music, such as community groups and religious settings, showed significant correlations and remained significant in the regression model. This suggests that broader cultural engagement relates to academic outcomes. Participation in organized arts settings can have a positive impact on strengthening discipline, confidence, persistence, and time management. These factors could possibly contribute to performance in school even if they are not activities directly related to academics.

The regression model explained 5.9% of GPA variance, while predictive models produced low R^2 values, ranging from negative to 0.04. This is one of the most important findings of the project. This indicates that music-related variables are statistically related to GPA, but explain only a small portion of overall performance. Academic performance is influenced by many variables not included in this dataset, such as household income, school quality, and teacher support.

Compared with these broader structural and psychological influences, music participation alone would not be expected to dominate GPA outcomes. With that said, the low predictive power of the models does not mean that music is unimportant or that it does not have an influence on children. It means that music is only one component within a much larger system, which shapes educational achievement.

The models also showed a tendency to cluster predictions near the mean GPA scores (around 0.80), a common issue when models have limited signal relative to data noise. In practical terms, the model could estimate average performance reasonably well but struggled to distinguish students at the low and high ends of the GPA distribution. This reinforces the conclusion that stronger predictors are needed for accurate forecasting.

The findings support a nuanced conclusion, backed by existing literature. Music education is not a strong predictor of academic achievement when measured in simple

survey terms alone. Music-related access and participation are associated with GPA in small but meaningful ways. The strongest and most stable relationship was the broader environment represented by access to instruments, not music lessons themselves. This suggests that music contributes to development as part of a system of opportunity, enrichment, and support. The question isn't whether music alone raises grades, but how music participation interacts with social and educational contexts to support overall child development. Viewing the impact of music in a broader way makes it an important subject worthy of continued research and educational investment.

4.1 Limitations

While this project provides useful evidence on the relationship between music education and academic performance, several limitations must be acknowledged. Recognizing these limitations was crucial for interpreting the current findings and guiding future research.

A primary limitation was the reliance on observational survey data rather than experimental data. Because students were not randomly assigned to receive music education, causation conclusions cannot be drawn. The analysis identified statistical associations between music-related variables and GPA, but could not determine whether music participation directly caused academic improvement.

A second limitation concerned the measurement of music participation itself. Many available PSID variables were binary yes/no indicators, such as whether a child took music lessons or participated in music activities. These measures did not capture important differences, such as the duration of participation, practice time, quality of instruction, ensemble participation, or consistency. As a result, impactful differences between students could not be observed and were compressed into simple categories.

The dependent variable used, GPA, was a useful but imperfect measure of academic performance. GPA varied across schools, districts, grading systems, and teacher expect-

tations. GPA normalization improved comparability across survey years, but grading practices were not fully standardized and could have reflected attendance, participation, or school policies.

The final limitation presented itself as the size of the dataset. Even with the original PSID datasets being substantially larger, observations with missing GPA values or incomplete predictor variables had to be removed for clean statistical modeling. This reduced the final regression sample and may have introduced some selection bias if students with missing data were systematically different from those retained in the dataset.

In summary, the limitations of this project did not weaken its central contribution. Rather, the limitations of this project clarify that the relationship between music education and academic performance is complex, context-dependent, and embedded in wider social structures.

4.2 Future Research

Despite the limitations of the project, it highlights promising directions for future research. Future studies should incorporate direct socioeconomic controls such as family income, parental education, and neighborhood measures. This would help in the separation of the effect of music participation itself from the effect of broader household advantage.

Another way to improve the strength of the observations could be through the usage of richer indicators of music engagement. Rather than binary participation measures, future datasets should capture factors such as the duration of study, practice frequency, ensemble participation, performance experience, and motivations. Such measures would provide a more realistic representation of music education.

Additionally, future work could examine other academic outcomes beyond GPA. Music may relate more strongly to standardized test scores, reading comprehension, at-

tention span, attendance, graduation rates, college enrollment, and more. Such outcomes could better reflect the kinds of skills music training is hypothesized to influence.

Utilizing more advanced predictive approaches such as random forests, gradient boosting, mixed-effects models, and more could show nonlinear relationships that were missed by the ordinary linear regression.

Finally, future research should also consider questions of equity and access. If music opportunities are associated with positive developmental outcomes, then unequal access to instruments, lessons, and school programs becomes an important issue on a higher level. Understanding who receives access to such resources could be as important as understanding the average effect of participation itself.

Future research that combines stronger controls, richer measures, and longitudinal designs can build on these findings and move closer to understanding how and who is most impacted by music education.

Appendix A

Additional Figures

This appendix includes supplemental figures for the separate yearly prediction models. These figures support the predictive modeling results discussed in the main text.

A.1 Yearly Prediction Figures

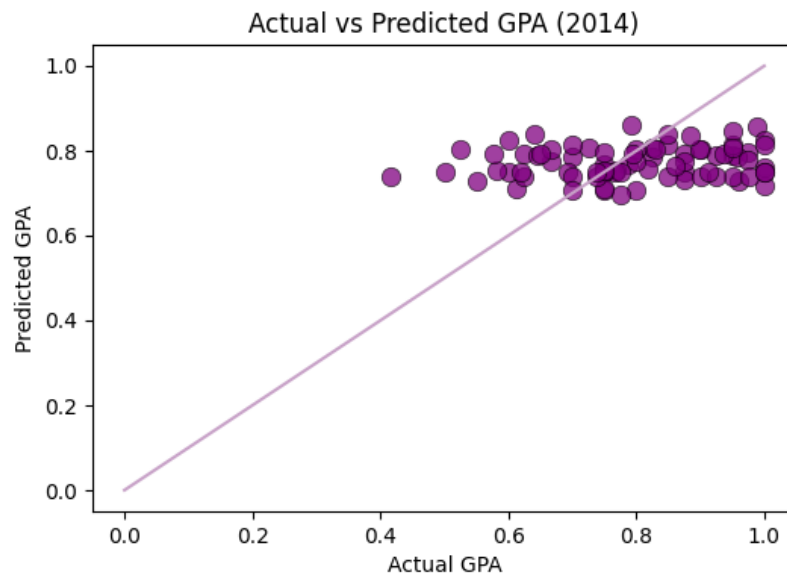


Figure A.1.1: Predicted versus actual GPA values for 2014

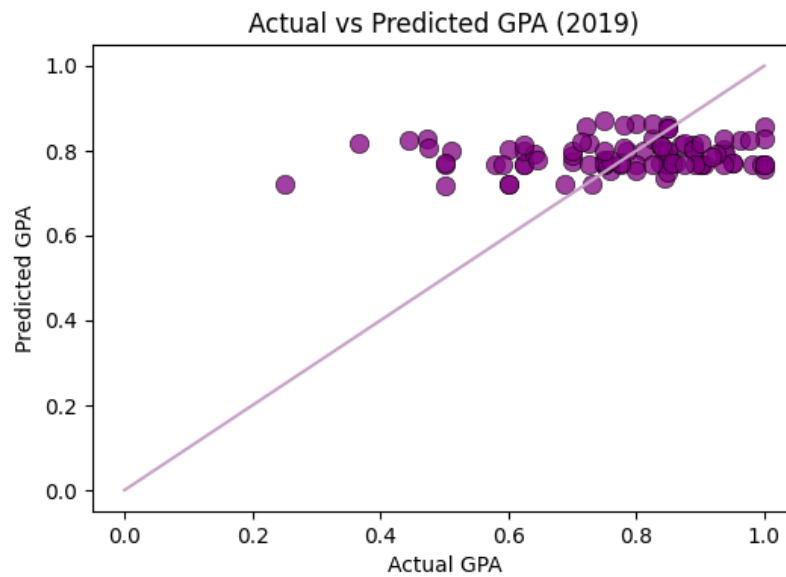


Figure A.1.2: Predicted versus actual GPA values for 2019

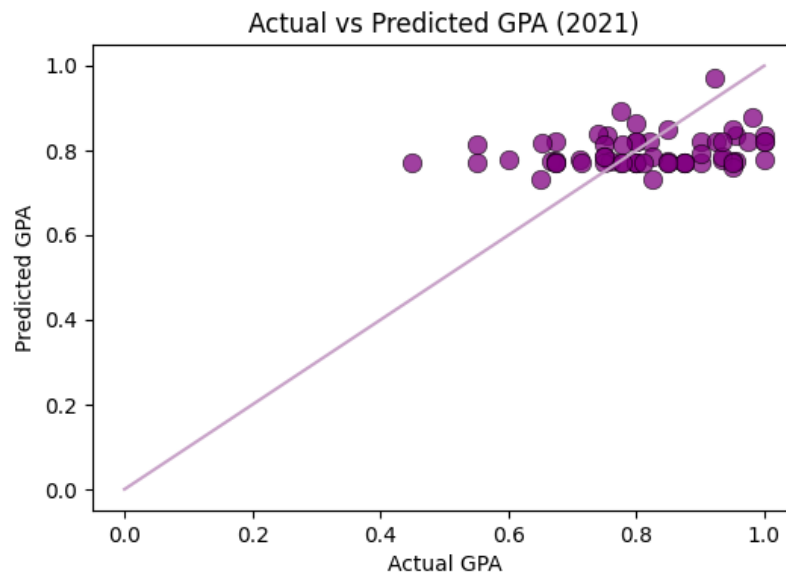


Figure A.1.3: Predicted versus actual GPA values for 2021

A.2 Yearly Music Lessons Participation Figures

The following figures show GPA distributions by music lesson participation across the three survey years.

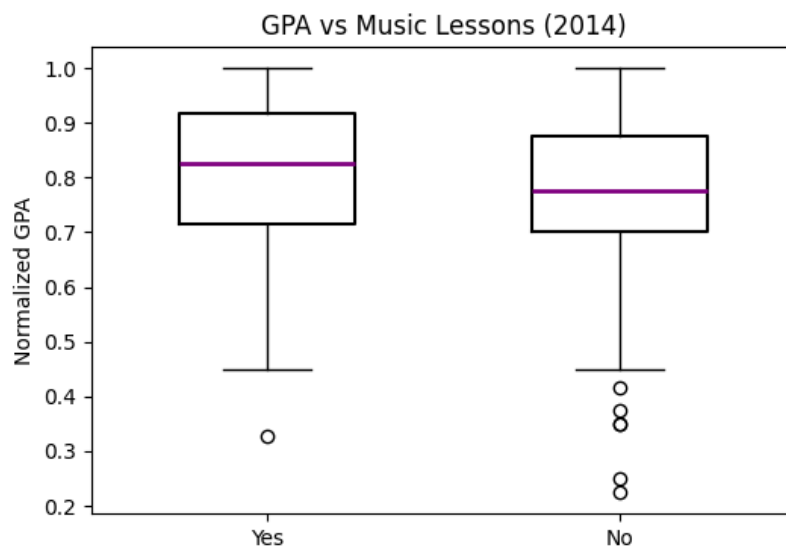


Figure A.2.1: GPA by music lesson participation in 2014

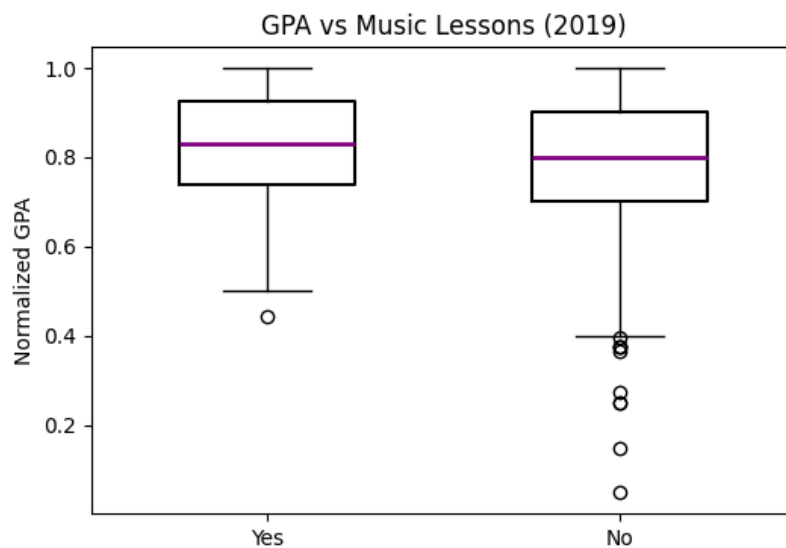


Figure A.2.2: GPA by music lesson participation in 2019

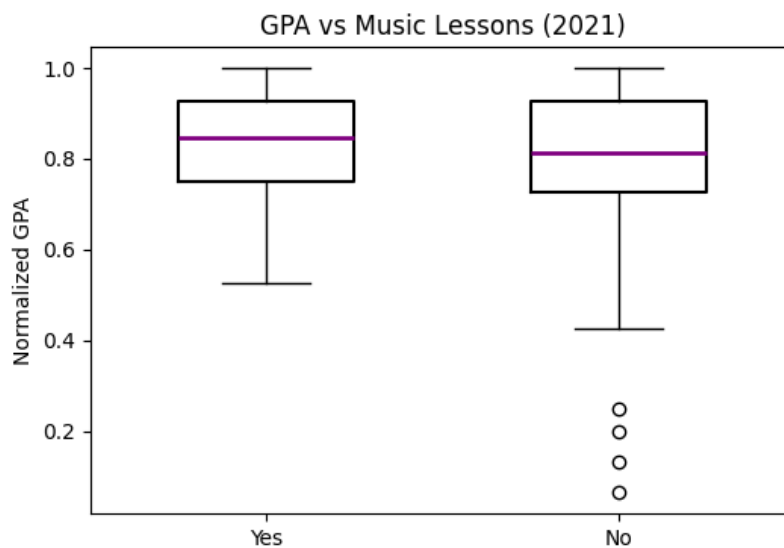


Figure A.2.3: GPA by music lesson participation in 2021

A.3 Yearly General Music Participation Figures

The following figures show GPA distributions by general music participation across the three survey years.

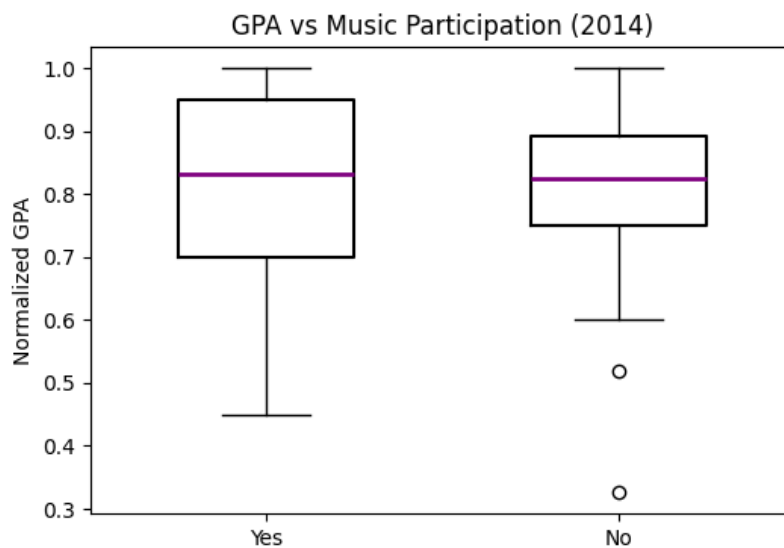


Figure A.3.1: GPA by general music participation in 2014

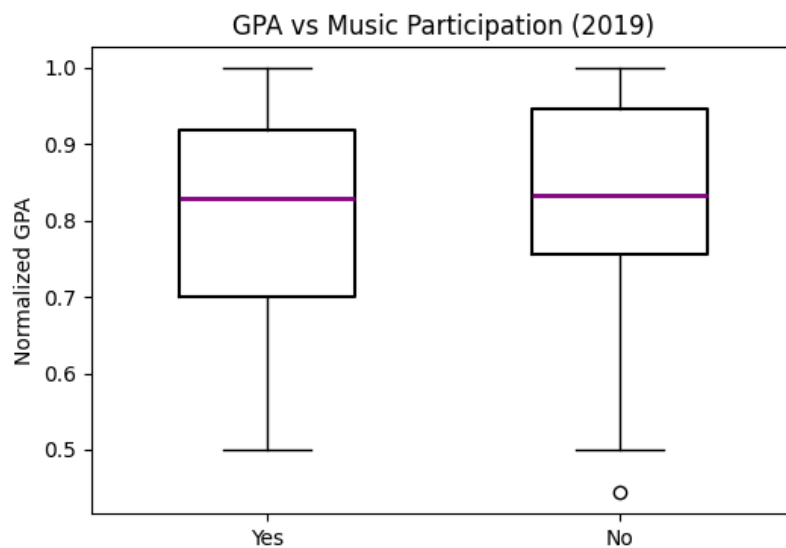


Figure A.3.2: GPA by general music participation in 2019

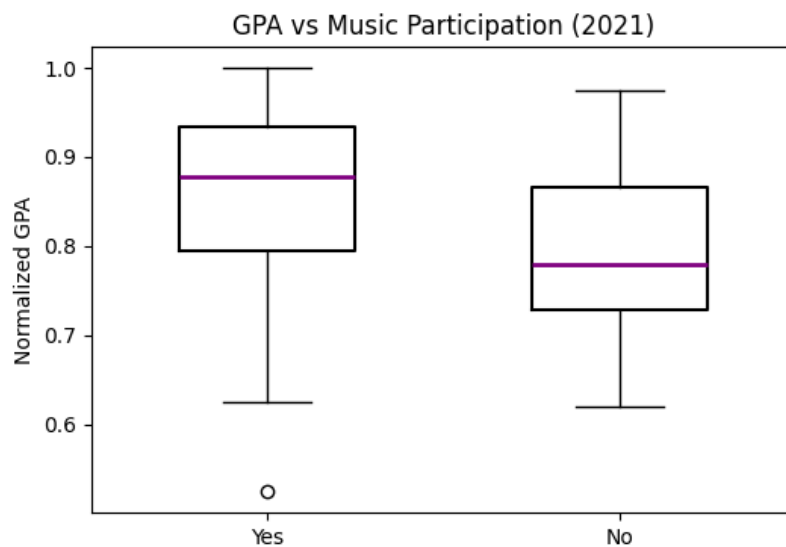


Figure A.3.3: GPA by general music participation in 2021

Appendix B

Code Appendix

B.1 Full Code Repository

The complete scripts for data processing, statistical analyses, and predictive modeling code used in this project are available in the following GitHub repository:

github.com/chikolov/SPROJ

The repository includes:

- Raw and cleaned datasets and their respective code-books used in the analysis
- Data cleaning and processing scripts
- Statistical analysis code (t-tests, correlations, regression)
- Predictive modeling implementation
- Code used to generate all figures for this project

This appendix includes selected code excerpts for reference.

B.2 Data Cleaning Example

```
import pandas as pd
```

```
df = pd.read_csv("2014.csv")

df["C14B12"] = pd.to_numeric(df["C14B12"], errors="coerce")
df["C14B13"] = pd.to_numeric(df["C14B13"], errors="coerce")

df = df[(df["C14B12"] > 0.01) & (df["C14B12"] <= 12)]
df = df[(df["C14B13"] > 0.01) & (df["C14B13"] <= 12)]

df["GPA_normalized"] = df["C14B12"] / df["C14B13"]

df.to_csv("cleaned_2014.csv", index=False)
```

B.3 Combined Dataset and Regression Example

```
import pandas as pd
import statsmodels.api as sm

data = pd.read_csv("combined_psid_music_data.csv")

variables = [
    "music_lessons",
    "music_participation",
    "instrument_home",
    "instrument_frequency",
    "electronics_music",
    "arts_tutoring",
    "arts_community",
```

```
"arts_religious",
"year",
"gpa_norm"
]

data = data[variables].dropna()
data = data[data["gpa_norm"] <= 1.0]

X = data.drop(columns=["gpa_norm"])
y = data["gpa_norm"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

B.4 Modeling Example

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```


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